# All-Angle Fixed Sets and Gaussian Koopman Spectrum for the Planar Harmonic Strobe

Route-A structural certificate C178


**Abstract**

For the planar oscillator $H = (q^2 + p^2)/2$, the time-$\theta$ strobe is a rotation. Its $n$th iterate fixes only the origin unless $n\theta \in 2\pi\mathbb{Z}$, when it fixes the whole plane. Thus irrational $\theta/(2\pi)$ gives the Artin–Mazur zeta $(1 - z)^{-1}$, whereas every rational angle has an uncountable resonant fixed set and no classical Artin–Mazur series. On invariant Gaussian $L^2$, a normalized Laguerre–angular basis has eigenvalues $e^{im\theta}$ with infinite radial multiplicity. Hence the Koopman unitary is noncompact, belongs to no finite Schatten class, and has no ordinary Fredholm determinant. No arithmetic or target-spectrum claim is made.



**Keywords:** harmonic oscillator; stroboscopic rotation; Gaussian Koopman spectrum; resonance; Artin–Mazur zeta; Route A.


## 1 Frozen flow and fixed sets

Hamilton's equations give

$$T_\theta(q,p) = (q\cos\theta + p\sin\theta, -q\sin\theta + p\cos\theta), \qquad T_\theta^n = T_{n\theta}.$$

A nonidentity rotation fixes only $(0,0)$, while the identity fixes $\mathbb{R}^2$. Therefore

$$\mathrm{Fix}(T_\theta^n) = \begin{cases} \mathbb{R}^2, & n\theta \in 2\pi\mathbb{Z}, \\ \{(0,0)\}, & n\theta \notin 2\pi\mathbb{Z}. \end{cases}$$

If $\theta/(2\pi)$ is irrational, every fixed count is one and

$$\zeta_{\mathrm{AM}}(z) = \exp\left(\sum_{n\geq 1} \frac{z^n}{n}\right) = \frac{1}{1-z}.$$

If $\theta/(2\pi) = a/b$ is reduced, the whole plane is fixed exactly when $b \mid n$. Its uncountable cardinality is not a finite Artin–Mazur coefficient, so the classical series is undefined. This includes the zero-angle edge $b = 1$.

The reflection $S(q,p) = (q,-p)$ is involutive and satisfies $ST_\theta S = T_\theta^{-1}$.

## 2 Gaussian Koopman diagonalization

Freeze $q - ip = re^{i\varphi}$ and

$$d\gamma = \pi^{-1}e^{-(q^2+p^2)}\,dq\,dp, \qquad U_\theta f = f \circ T_\theta.$$

Rotation preserves $\gamma$ and advances $\varphi$ by $\theta$. For $m \in \mathbb{Z}$ and $k \geq 0$, set

$$\psi_{k,m} = \sqrt{\frac{k!}{(k+|m|)!}}\, r^{|m|} L_k^{|m|}(r^2) e^{im\varphi}.$$

Angular Fourier completeness and the generalized-Laguerre identity

$$\int_0^\infty e^{-x} x^{|m|} L_k^{|m|}(x) L_\ell^{|m|}(x)\,dx = \frac{(k+|m|)!}{k!}\delta_{k\ell}$$

prove that these functions form an orthonormal basis. Moreover

$$U_\theta \psi_{k,m} = e^{im\theta}\psi_{k,m}.$$

For irrational $\theta/(2\pi)$, the eigenvalues indexed by $m$ are distinct and dense on the unit circle. For reduced rational $a/b$, they are exactly the $b$th roots of unity. In either case, the unrestricted radial index $k$ gives every eigenvalue infinite multiplicity. Thus $U_\theta$ is noncompact, is in no finite Schatten class, and has no ordinary trace-class determinant $\det(I - zU_\theta)$ for $z \neq 0$.

With $V_S f = f \circ S$ and complex conjugation $K$, the antiunitary $\Theta_G = V_S K$ satisfies $\Theta_G U_\theta \Theta_G^{-1} = U_\theta^{-1}$.

## 3 Decision boundary

The finite ledger supplied with the paper is a regression sentinel for the rotation formula, not a proof. The result uses no target zero or prime table, fitted parameter, arithmetic local datum, Euler factor, root number, automorphy object, or Hilbert–Pólya assertion. The scope is `NO_BAD_EULER_OR_ROOT_NUMBER`, and Route B is unauthorized.

**Declarations.** All data and code are included. No external dataset, private data, or human or animal subject is involved. The anonymous contribution covers conceptualization, proof, software, validation, and writing. No funding or competing interest is declared. AI assistance was used for drafting and code generation; claim-bearing formulas were separately reconstructed by distinct internal code paths, and no external reviewer or acceptance score is represented.